

LEAN POST-TYPHOON RECOVERY: MULTI-AGENT LOOK-AHEAD AND CONSTRAINTS FOR THE LAST PLANNER SYSTEM

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ABSTRACT

Post-typhoon recovery involves high-variability work in which plan reliability is frequently undermined by late discovery of constraints, fragmented information, and rework. Rapid post-typhoon building recovery requires integrating heterogeneous evidence and converting it into actionable and auditable plans. This paper proposes a Lean Construction informed, knowledge-driven framework that combines a multi-modal knowledge graph (MMKG) with multi-agent orchestration for end-to-end building damage assessment and recovery planning, aligned with the Last Planner System (LPS) through look-ahead planning, make-ready checks, and explicit constraint management. The MMKG capture emergency management criteria, typhoon evolution and exposure status, building-level damage evidence, and reusable post-disaster recovery templates. Each evidence node and key relationship stores source, spatiotemporal scope, confidence level, and version metadata, thus linking each conclusion to a traceable chain of evidence and enabling querying and answering questions. The approach uses a shared knowledge base as the single source of truth, enabling multi-agent system to generate a constraint log, perform readiness verification, and release work only when prerequisites are satisfied, thereby reducing avoidable waiting and rework.

KEYWORDS

Last Planner System, Multi-agent, Knowledge-driven, Multi-modal knowledge graph, Typhoon disaster

INTRODUCTION

Typhoons are intense tropical cyclones that form in the Northwest Pacific Ocean and its adjacent waters, and are among the most destructive natural disasters globally. Their passage is typically accompanied by a complex process of destructive winds, heavy rainfall, flooding, and storm surges, which can cause systemic impacts on building structural safety, building envelopes, and functionality, further amplifying the risks to infrastructure and residential areas in coastal and low-lying regions (Shan et al., 2023).

After typhoons and other tropical cyclones make landfall, emergency command centers need to assess building damage and produce repair and recovery plans within a short period, decisions that can impact lives and property. However, a large amount of information emerges in different modalities after a disaster and is frequently updated, making it difficult

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for even professional staff to fully grasp it immediately (Zhou et al., 2018). For example, while aerial photographs or satellite images can quickly reveal the state of damage, they lack unified semantic alignment with government loss statistics, typhoon paths and weather patterns, and on-site disaster reports, leading to slow cross-departmental collaboration and difficulties in tracing evidence chains (Koldasbayeva et al., 2024). Beyond information integration, a critical operational bottleneck is that recovery planning becomes unstable when constraints are discovered late, causing rework and unreliable commitments under time pressure.

From a Lean Construction perspective, post-typhoon recovery can be viewed as a high-variability production system in which plan reliability depends on reducing workflow variability, making constraints explicit, and enabling auditable commitments (Garcés et al., 2025). The Last Planner System (LPS), as a Lean-based production planning and control method, emphasizes look-ahead planning, make-ready processes, and weekly work planning, where tasks should be released only when prerequisites are satisfied and constraints are actively tracked (Ballard, 2000). Through a case study of Typhoon Yagi (No. 11 of 2024) in a representative region, the case study demonstrates multi-source evidence alignment, conflict detection, and rule-based make-ready checking, and illustrates how evidence-linked planning artifacts can improve transparency and support cross-departmental coordination under uncertainty. Motivated by this, we propose a Lean-oriented, knowledge-driven framework for post-typhoon building recovery that directly supports LPS look-ahead and constraint management.

LITERATURE REVIEW

MMKG FOR DISASTER AND DECISION-MAKING

The core challenge of post-disaster management lies not only in damage identification, but also in ensuring that visual evidence, statistical tables, path and meteorological time series, and textual disaster reports are aligned, queryable, and auditable within the same semantic and spatiotemporal framework. MMKG provide a path to unify heterogeneous data into entities, relationships, events, and spatiotemporal constraints, enabling cross-departmental information collaboration to shift from manual information collection to structured retrieval and storage (Song et al., 2024).

In disaster scenarios, the combination of Knowledge Graph (KG) and LLM is becoming an important direction for controlled generation and evidence-based reasoning. Chen et al. (Chen et al., 2024) proposed constructing an emergency knowledge graph and guiding LLM reasoning on the graph with chained prompts, directly addressing the problems of hallucinations and unstable reasoning by introducing external structured constraints, and emphasizing evidence-driven decision output in multi-stage emergency tasks. For the typhoon domain, Huang et al. (Huang et al., 2026) proposed TyphoonKG, further demonstrating a route for end-to-end typhoon knowledge graph construction and application using an LLM-agent, emphasizing hierarchical knowledge representation and dynamic updates to support management tasks such as trajectory tracking and impact analysis.

MULTI-AGENT ORCHESTRATION FOR LEAN POST-TYPHOON RECOVERY PLANNING

Multi-agent systems (MAS) provide a structured paradigm in which specialized agents collaborate to complete complex tasks (Li et al., 2024). In recent years, large language models have increasingly served as the interaction and reasoning core for such systems, supporting planning, execution, reflection, and remediation through message passing. For post-typhoon recovery, the primary challenge is not only knowledge access but also production control under high variability. Recovery activities must be translated into executable work with explicit prerequisites, responsibilities, and verifiable evidence.

Recent studies have explored LLM-agent approaches for improving typhoon knowledge foundations and mitigating error accumulation in spatiotemporal modeling (Huang et al., 2026), indicating a trend toward multi-agent assisted knowledge construction. Nevertheless, much of the existing work emphasizes knowledge organization and query services, while offering limited support for Lean-oriented recovery orchestration, especially for look-ahead planning, make-ready checks, and explicit constraint management required by the Last Planner System (LPS) (Soman & Molina-Solana, 2022). In post-disaster contexts, recovery requires multi-step, cross-modal, and verifiable reasoning chains that directly serve planning and control, rather than producing isolated summaries or recommendations. This motivates a problem-driven integration of multi-agent orchestration with graph-based constraints to enable evidence-linked, auditable, and executable recovery planning (Xu et al., 2025).

RESEARCH GAP

Based on the above, the gaps in publishable research on post-typhoon building damage and recovery, from a Lean Construction and LPS-oriented perspective, can be summarized as follows:

1. Lack of a single source of truth for LPS look-ahead and constraint logs. Multi-source data such as typhoon tracks and meteorological timelines, loss statistics, image evidence, and text reports often lack unified entity identifiers, spatiotemporal references, and provenance links. This prevents consistent constraint identification and readiness verification, leading to avoidable waiting, duplicated processing, and rework during cross-departmental collaboration.
2. Insufficient planning reliability evaluation for the recovery phase under high variability. Existing geographic data-driven modeling faces systematic issues such as bias, imbalance, spatial autocorrelation, and uncertainty, which can cause unstable performance in cross-event and cross-region transfer if not explicitly addressed. Moreover, MAS without explicit verification and safety constraints may produce untraceable error chains, undermining the reliability of commitments and the audibility required for Lean planning and control (Koldasbayeva et al., 2024; Wang et al., 2025). Limited practical methodologies for multi-objective trade-offs and multi-risk coupling that are integrated into LPS routines. The recovery phase involves competing objectives and coupled risks, yet existing work often lacks actionable mechanisms that connect evidence governance to executable planning routines, including rolling look-ahead updates, make-ready checks, and rule-based task release under regulatory and safety constraints. Reviews on recovery planning tools and indicators highlight these limitations, and LLM-focused disaster management reviews also indicate insufficient coverage of the recovery phase and production-control needs (Mohammadi et al., 2024).

METHODOLOGY

Based on the shortcomings in post-typhoon building management, this paper formulates post-typhoon recovery as a Lean production control problem and designs an end-to-end workflow aligned with the LPS. The core idea is to use a MMKG as a single source of truth for evidence governance and constraint traceability, and to operationalize LPS routines through multi-agent orchestration, including rolling look-ahead planning, explicit constraint management, make-ready verification, and rule-based task release to a weekly work plan, as shown in Figure 1. The method consists of two major components. First, we construct an MMKG that unifies and aligns multi-source information such as post-event imagery, loss statistics, typhoon tracks and meteorological sequences, and field reports, while explicitly recording provenance, spatiotemporal scope, confidence, and versioning to support auditable planning. Second, we design a multi-agent workflow that produces LPS planning artifacts rather than one-shot conclusions. MAS collaboratively generate a look-ahead plan, maintain a living constraint log, perform make-ready checks using evidence-linked rules, and release only

executable tasks to the weekly work plan, enabling reliable commitments under high variability.

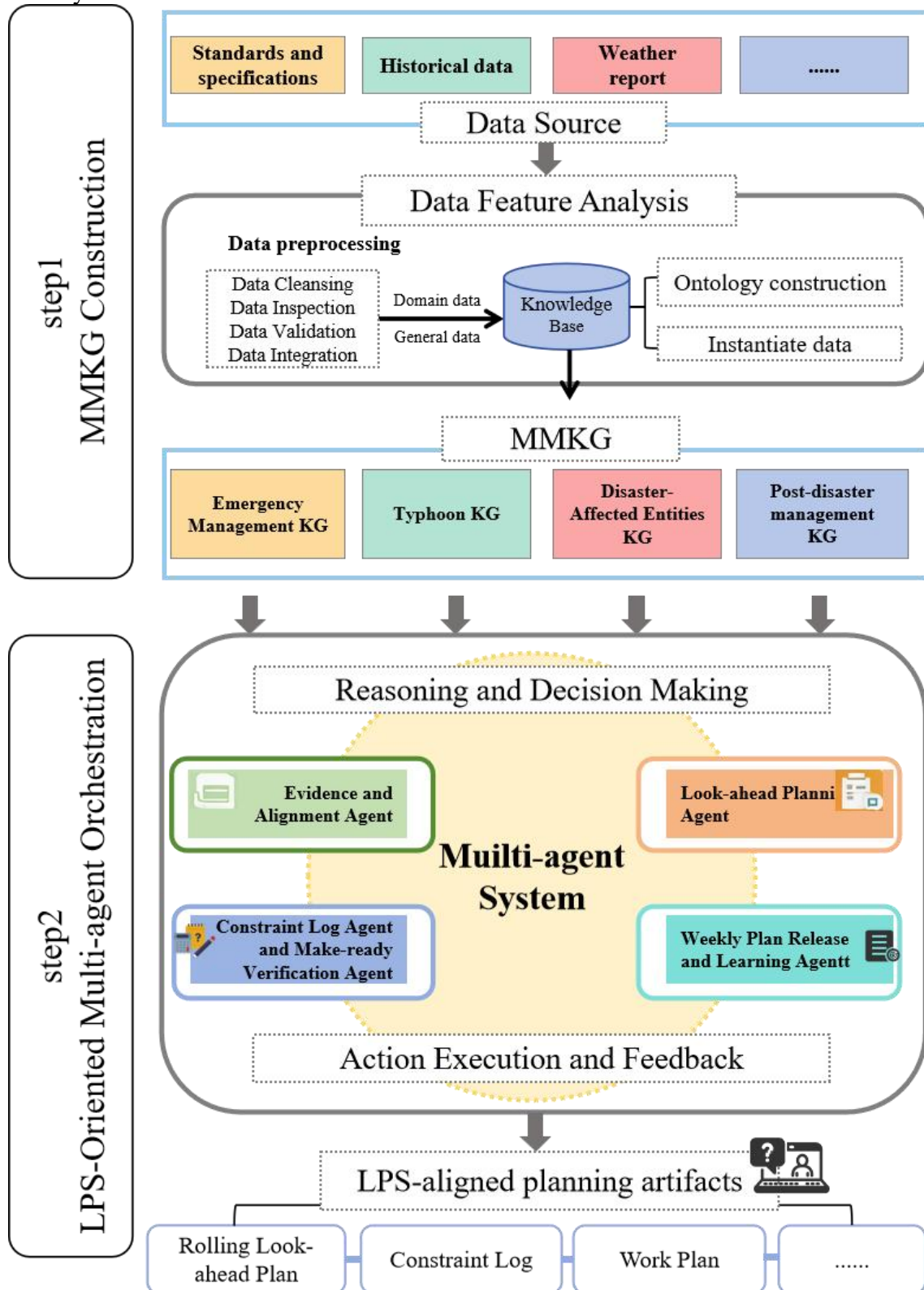


Figure 1: Framework of Methodology

MMKG CONSTRUCTION

Knowledge representation is the foundation and prerequisite for knowledge application. Understanding and reconstructing the typhoon-related knowledge framework is crucial for

transforming widespread unstructured knowledge into practical structured resources. Based on existing typhoon disaster research (Chen et al., 2024; Oktari et al., 2020). This paper establishes a method for representing typhoon-related knowledge from the perspectives of emergency management regulations, typhoon information, building information, and post-disaster recovery.

Emergency Management Standards KG

The role of the emergency management normative framework is to transform institutionalized norms into searchable and executable structured knowledge, thereby enabling multi-agent task decomposition and decision generation to have boundaries and auditability. During construction, normative clauses can be abstracted into a minimal semantic skeleton of stage — subject — action — condition — evidence — responsibility . The emergency management standards KG transforms institutional rules into executable planning constraints for LPS. Normative clauses are abstracted into a minimal structure that supports make-ready verification, including stage, responsible party, permitted action, prerequisite condition, required evidence, and acceptance criteria. These clauses define feasibility boundaries and compliance constraints for task release, and enable auditable commitments by linking each planned action to explicit prerequisites and evidence.

Typhoon KG

The typhoon knowledge graph serves as the main representation of the event evolution and impact chain, unifying multi-source post-disaster data into a single spatiotemporal reference. Following the construction approach of TyphoonKG (Huang et al., 2026). Relevant event fragments from disaster reports are extracted and instantiated into the KG, retaining their source and confidence level to support subsequent conflict interpretation and dynamic updates. To enable multi-agent orchestration, the graph should also explicitly store available retrieval anchors (such as events, regions, time windows, impact thresholds, etc.) and traceable evidence paths for each task.

Disaster-affected Entities KG

Disaster-Affected Entities KG is to address the difficulty of translating assessment results into actionable restoration targets. The key is to treat buildings as disaster-bearing entities, aligning imagery, statistics, and textual evidence at the building scale, and linking damage typification and restoration action templates to the same semantic chain. During construction, the Building (such as including location, footprint, usage/structure type, and criticality level, etc.) serves as the core entity, connecting ImageEvidence (such as post-disaster imagery and visual model output , etc.), LossRecord (such as county or street-level statistical losses, etc.), and post-disaster building damage for assessment (Adriano et al., 2021).

Post-disaster Management KG

The post-disaster management KG provides reusable recovery routines and control indicators consistent with Lean planning and improvement. It encodes recovery phases, task templates, handoff conditions, risk coupling, and uncertainty disclosures as queryable structures. The graph also stores planning-control indicators and diagnostic categories such as reasons for non-completion, enabling continuous improvement loops that refine constraint recognition and make-ready rules across events.

LPS-ORIENTED MULTI-AGENT ORCHESTRATION

Given the MMKG as a shared memory and constraint foundation, the multi-agent system is designed to operationalize LPS routines rather than generate one-shot recommendations (Cheng et al., 2024). Using knowledge in the MMKG as constraints, the MAS decomposes

the objective into four types of tasks: information aggregation, conflict diagnosis, solution generation, and solution verification. The workflow is organized around four LPS-centered functions.

1. Evidence and Alignment Agent consolidate cross-modal inputs and maintain the MMKG as a single source of truth, producing evidence-linked building states and flagging conflicts with provenance and confidence annotations.

2. Look-ahead Planning Agent generates a rolling lookahead plan by decomposing recovery objectives into candidate work packages and sequencing them by risk, accessibility, and dependencies retrieved from the MMKG.

3. Constraint Log Agent and Make-ready Verification Agent explicitly identify prerequisites for each candidate task, record them in a living constraint log with responsible parties and due conditions, and perform make-ready checks using standards KG rules and evidence links. Tasks that fail readiness checks are not released and are returned to evidence alignment or constraint resolution steps.

4. Weekly Plan Release and Learning Agent releases only executable tasks to a weekly work plan based on readiness status and resource constraints. During execution, it records completion-to-commitment outcomes and reasons for non-completion, and writes back validated changes and diagnostic signals to the MMKG, enabling iterative improvement of constraint identification and planning rules.

The final outputs are LPS-aligned planning artifacts with traceable evidence, including a rolling look-ahead plan, a constraint log, make-ready status for each task, and a weekly work plan with evidence-linked commitments. Damage lists and recovery priorities are treated as intermediate products that support planning, and all released tasks retain auditable evidence paths and explicit constraint states.

CASE STUDY

EVENT AND DATASET

Yagi (No. 11 of 2024) as a case study, selecting some buildings in a certain region as samples of damaged structures, typhoon information shows in table 1,2 and buildings information shows in table 3. After landfall, forming a persistent chain of strong winds, rain, and complex disaster-causing factors. Yagi as a significant autumn typhoon landfall, characterized by high intensity, wide impact, dispersed post-disaster information sources, and significant pressure on inter-departmental coordination. Therefore, it is suitable as a representative scenario for "multi-source evidence alignment and executable recovery orchestration."

The dataset is constructed around an end-to-end chain of post-disaster building damage assessment and recovery orchestration, containing five modalities uniformly mapped to MMKG entities. The first category is post-disaster image evidence, used to generate building-level DamageObservations, recording imaging time, coverage area, and model confidence. The second category is a loss statistics table, including indicators such as the number of damaged items, damage type, and damage assessment, serving as a macro-level constraint and consistency reference beyond building-level evidence. The third category is typhoon path and intensity sequences, using meteorological department reports or optimal path products, using path points and time windows as spatiotemporal anchors for exposure inference. The fourth category is disaster report texts, emergency management regulations, and related standards, broken down into locatable ReportClaims (location, object, damage, demand, source paragraphs), retaining metadata tags to support conflict interpretation and verification. The fifth category is general reference cases for post-disaster recovery, including historical data and cases on how personnel and resources were allocated for post-disaster management and building reconstruction.

Table 1: Yagi (No. 11 of 2024) Summary Information

Storm ID	Start	End	Max Intensity
2024244N09137	Aug 30 12Z	Sep 9 12Z	145 kt (Sep 5 18Z), 907 mb (Sep 5 18Z)

Table 2: Part of Yagi (No. 11 of 2024) Position and Intensity

Time	Wind Speed	Center Position	Intensity
2026/9/4 8:00	35m/s	19.0N/117.7E	Typhoon
2026/9/4 9:00	35m/s	19.0N/117.7E	Typhoon
2026/9/4 10:00	35m/s	19.0N/117.7E	Typhoon
2026/9/4 11:00	38m/s	19.1N/117.6E	Typhoon
2026/9/4 13:00	38m/s	19.1N/117.5E	Typhoon
2026/9/4 14:00	42m/s	19.1N/117.4E	Strong typhoon
2026/9/4 15:00	42m/s	19.1N/117.3E	Strong typhoon
2026/9/4 16:00	42m/s	19.1N/117.2E	Strong typhoon
2026/9/4 17:00	48m/s	19.1N/117.2E	Strong typhoon
2026/9/4 18:00	48m/s	19.2N/117.1E	Strong typhoon
2026/9/4 19:00	48m/s	19.1N/116.8E	Strong typhoon
2026/9/4 20:00	50m/s	19.1N/116.7E	Strong typhoon
2026/9/4 21:00	50m/s	19.1N/116.6E	Strong typhoon
2026/9/4 22:00	50m/s	19.1N/116.5E	Strong typhoon
2026/9/4 23:00	52m/s	19.1N/116.3E	Super typhoon
2026/9/5 0:00	52m/s	19.1N/116.3E	Super typhoon
2026/9/5 1:00	52m/s	19.1N/116.2E	Super typhoon
2026/9/5 2:00	55m/s	19.1N/116.0E	Super typhoon
2026/9/5 3:00	55m/s	19.2N/116.0E	Super typhoon
2026/9/5 4:00	55m/s	19.2N/116.0E	Super typhoon

Table 3: Part of buildings information

Classify	Picture	Time	Position	Damage Situation
Steel structure		2024/9/10 16:00	110.039996N/ 19.937942E	Severe damage
Steel structure		2024/9/10 16:00	110.039275N/ 19.940831E	Local damage
Membrane structure		2024/9/10 16:00	110.287286N/ 20.041453E	Local damage
Membrane structure		2024/9/10 16:00	110.496389N/ 19.970091E	Local damage

IMPLEMENTATION DETAILS

MMKG schema for LPS-oriented planning

MMKG is adopting a shared core entity and sub-graph extension design, the schema centralizes eight core entity types: Event (typhoon event), Region (administrative region or grid), Building, TimeWindow, Evidence, Action (recovery action), Rule (standard rule), and Resource. Building serves as the focal object for post-disaster management, strongly linked to Evidence, DamageAssessment, and RecoveryPlan, while Event anchors the spatiotemporal framework through strong associations with Region and TimeWindow. Each sub-graph emphasizes distinct aspects: the normative KG defines stage-based responsibility chains, triggering conditions, and mandatory constraints; the typhoon KG provides sequences of path points, intensity evolution, and impact scopes; the building KG supplies post-disaster image evidence, damage levels, statistical records, and geographic locations; and the general post-disaster KG offers multi-objective trade-offs, risk representations, and reusable recovery process templates to ensure cross event applicability.

For multi-source data ingestion and information extraction, structured image evidence objects are first generated from post-disaster imagery via visual models or rule-based methods, containing building candidate boxes, damage segmentation results, damage-level probabilities, and quality labels, all stored as ImageEvidence nodes. Loss statistics tables are directly mapped to LossRecord nodes with field-level alignment to Region via region_id. Disaster reports are processed through a chained pipeline of stepwise extraction and alignment: initially extracting event fragments (location, time, object, damage description, and handling actions), then linking entities to Building, Region, and finally binding sentence-level evidence to structured entries to form query-able evidence paths. A fundamental engineering requirement for the MMKG is to bind every conclusion to a traceable evidence chain, as shown in Figure 2.

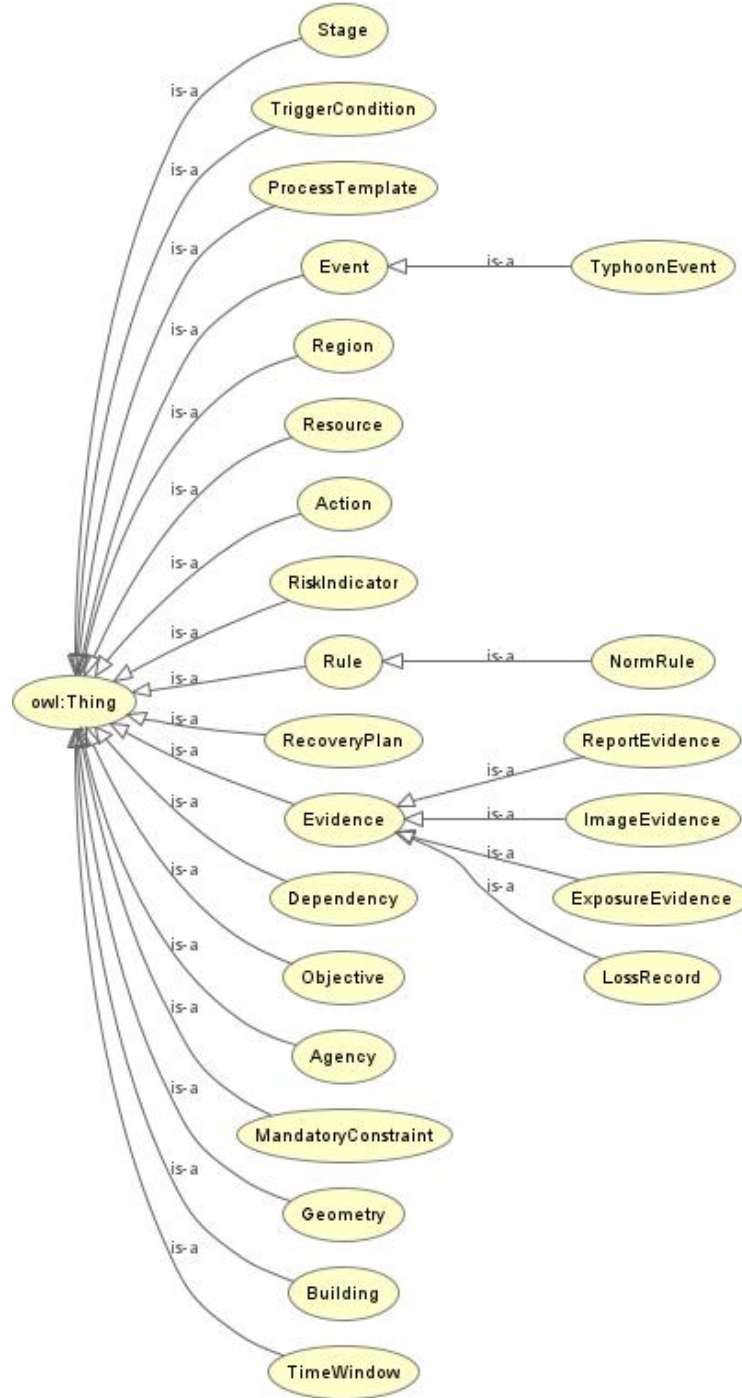


Figure 2: Yagi Typhoon Ontology

LPS-Oriented Multi-Agent Implementation

The MAS uses the MMKG as its shared memory and constraint foundation, forming a closed loop around the four capabilities you proposed: conflict detection, intelligent review, simulation generation, and verification and repair. The system consists of several agent roles, and an orchestrator controls the iterative process of query-write-requery to prevent the uncontrollable spread of errors caused by free generation. The multi-agent orchestration references the "plan, execute, reflect, correct" paradigm commonly used in agent research, with the core being the breakdown of complex processes into verifiable subtasks, using the graph query results as input for each step.

MAS down the objective into task chains and query templates. A query example illustrated in Figure 3.

Review agent	Design agent	Simulation agent	Calibration agent
<pre> SELECT ?b ?b_id ?geom ?critical ?imgDamage ?imgScore ?imgProv ?lossMetric ?lossProv ?reportClaim ?reportSpan ?reportProv ?exposureWind ?exposureRain ?exposurePr ov WHERE { ?e a ex:TyphoonEvent ; ex:eventId "event_id" . ?region a ex:Region ; ex:adminName "position" . ?b a ex:Building ; ex:buildingId ?b_id ; ex:locatedIn ?region ; ex:geometry ?geom ; ex:criticality ?critical . } </pre>	<pre> SELECT ?damageType ?actionTpl ?stage ?trigger ?f orbid ?requireEvidence ?ruleProv WHERE { ?rule a ex:NormRule ; ex:appliesToDamage ?damageType ; ex:stage ?stage ; ex:requiredEvidenceType ?requireEvidence ; ex:provenance ?ruleProv . } </pre>	<pre> SELECT ?plan ?b ?actionInst ?actionTpl ?priorit y WHERE { ?plan a ex:PlanCandidate . ?plan ex:hasAction ?actionInst . ?actionInst ex:forBuilding ?b ; ex:priority ?priority . } ORDER BY DESC(?priority) </pre>	<pre> SELECT ?b ?imgDamage ?reportDamage ?lossMet ric ?windIdx ?rainIdx ?conflictType ?provPath WHERE { ?b a ex:Building . OPTIONAL { ?img a ex:ImageEvidence ; ex:aboutBuilding ?b ; ex:damageLevel ?imgDamage ; ex:provenance ?p1 . } FILTER(bound(?imgDamage) && bound(?reportDamage) && (?imgDamage != ?reportDamage)) BIND("image_vs_report" AS ?conflictType) BIND (CONCAT(str(?p1)," ",str(?p2)," ",str(?p3)," ",str(?p 4)) AS ?provPath) } </pre>

Figure 3: Part of Yagi Typhoon query example

1. Evidence and Alignment Agents retrieve and consolidate cross-modal inputs into an evidence pack per building and per region, update MMKG entries, and flag conflicts with provenance and confidence annotations. Conflicts include inconsistency between imagery-derived damage and textual claims, mismatch between statistics definitions and aggregated building states, and implausible exposure-damage patterns.

2. Lookahead Planning Agent decomposes recovery objectives into candidate work packages and generates a rolling lookahead plan by sequencing tasks according to exposure severity, accessibility, dependencies, and criticality retrieved from MMKG.

3. Constraint Log and Make-Ready Verification Agent identifies prerequisites for each candidate work package, records them as a living constraint log with responsible parties and due conditions, and performs make-ready checks using standards KG rules and evidence links. Tasks that fail readiness checks are not released and are returned to evidence supplementation, alignment correction, or constraint resolution steps.

4. Weekly Plan Release and Learning Agent releases only executable tasks to a weekly work plan based on readiness status and resource constraints. During execution, it records completion-to-commitment outcomes and reasons for non-completion, and writes back validated changes and diagnostic signals to MMKG, enabling iterative improvement of constraint identification and planning rules.

CONCLUSION

This paper presents a Lean-oriented framework for post-typhoon building recovery planning and control by integrating a MMKG with an LPS-oriented multi-agent orchestration. By unifying post-disaster imagery, loss statistics, typhoon tracks and meteorological sequences, and textual reports into a single source of truth, the MMKG supports consistent spatiotemporal alignment, provenance tracking, and auditable constraint checking. Building on this foundation, the proposed orchestration operationalizes key Last Planner System (LPS) routines rather than producing one-shot recommendations. It generates a rolling look-ahead plan, maintains a living constraint log with responsibilities and due conditions, performs make-ready verification based on standards and evidence links, and releases only executable tasks into a weekly work plan with traceable commitments. The Typhoon Yagi case study demonstrates the feasibility of producing LPS planning artifacts and explainable recovery decisions under multi-source uncertainty, while reducing late constraint discovery and re-planning through evidence-linked verification.

Future work will focus on scaling the approach to city-scale building inventories and multi-agency coordination, improving the representation and propagation of uncertainty in evidence and constraints for risk-aware task release, and integrating real-time data streams to

support continuous look-ahead updates and learning from reasons for non-completion. These directions aim to strengthen plan reliability and transparency in high-variability disaster recovery settings, consistent with Lean Construction principles of reliable workflow and explicit constraint removal.

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